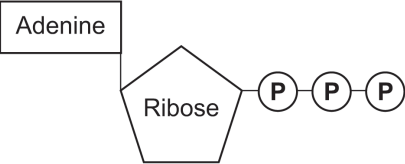


3.1.6 ATP

Content	Opportunities for skills development
A single molecule of adenosine triphosphate (ATP) is a nucleotide derivative and is formed from a molecule of ribose, a molecule of adenine and three phosphate groups.  Hydrolysis of ATP to adenosine diphosphate (ADP) and an inorganic phosphate group (P_i) is catalysed by the enzyme ATP hydrolase. • The hydrolysis of ATP can be coupled to energy-requiring reactions within cells. • The inorganic phosphate released during the hydrolysis of ATP can be used to phosphorylate other compounds, often making them more reactive. ATP is resynthesised by the condensation of ADP and P_i. This reaction is catalysed by the enzyme ATP synthase during photosynthesis, or during respiration.	